

CoreLog Vision: Lithology from Drill-Core Imagery, with Sim-to-Real Out-of-Distribution Detection

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Technical report · v1.2, 18 September 2026 · Licence CC BY 4.0
 CAOS open-research programme · CAOS_CoreLog, lithology from drill-core imagery
 Version DOI 10.5281/zenodo.22834528 · concept DOI (always latest) 10.5281/zenodo.21519229
 Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_CoreLog

Abstract. Logging the lithology of drill core, identifying the rock type down a borehole from the core in its trays, is slow and subjective manual work, and automating it from core-tray photographs is a natural target for computer vision. Labelled core photographs are scarce, so a model is often trained on synthetic core or on one operation’s core and then applied to another’s, where it answers confidently and wrongly unless it can tell that the new core is unlike its training data. CoreLog Vision classifies core-tray imagery into lithologies and detects core that lies outside its training distribution. On synthetic core a small convolutional network reaches 99.4% accuracy over six lithologies, against 92.9% for a colour-and-texture baseline, on a split that holds out entire boreholes. On real core from the open DCID-7 drill-core image dataset, a linear head over frozen ImageNet MobileNetV3-Small features reaches 99.2% over seven lithologies, and a linear head over the features of the synthetic-trained network reaches 88.8%; the two heads differ in backbone and in input resolution, so the difference does not isolate the cost of synthetic training. Nine out-of-distribution detectors are benchmarked at separating real DCID-7 core from held-out synthetic core, with both streams reduced to the same 24-pixel window. Distance detectors in a network’s feature space reach an area under the precision-recall curve (AUPR) of 0.92 to 0.999, softmax-based scores 0.74 to 0.83, and the reconstruction error of an autoencoder ranks the two populations in the wrong order (AUPR 0.28, below the chance level of 0.38). Distance in a feature space detects this shift; reconstruction error does not.

Index Terms. Drill-core logging, lithology classification, computer vision, convolutional neural networks, out-of-distribution detection, sim-to-real, Mahalanobis distance, mineral exploration, distribution shift.

I. INTRODUCTION

A Geologist logs a drill hole by reading its core, tray by tray, and naming the rock type at each depth, a task that is slow, tiring and subjective, and that produces the lithology column every downstream mine-planning decision rests on. Photographs of the core trays are routinely taken, so classifying lithology from those images with computer vision is a natural automation, and convolutional networks do

it well [1], [2], [4], [5]. The obstacle is the data. Labelled core imagery is scarce and operation-specific: core from one deposit, one drill and one photographic setup looks different from another’s, so a model trained on synthetic core or on one mine’s photographs faces a distribution shift when applied elsewhere. A classifier answers confidently on such out-of-distribution core and can be wrong without any sign of it, which in a logging context means a corrupted lithology column that no one flagged.

CoreLog Vision segments a core-tray image along each core channel, classifies each window’s lithology with a small convolutional network run in the browser, and stitches a depth log. Two further elements address the problem above. The classifier is evaluated on a split that holds out entire boreholes, so the accuracy is that of a new hole rather than of patches from holes seen in training. And an out-of-distribution detector flags incoming core that is unlike the training data. The classifier is evaluated on synthetic and on real core, and nine detectors are benchmarked on the sim-to-real shift.

II. THE LITHOLOGY CLASSIFIER

The classifier is a small convolutional network, two convolutional layers of 16 and 32 channels with max pooling, a 64-unit feature layer and a six-way softmax, applied to 24×24 -pixel RGB windows along each core channel. It is trained on procedurally generated core in six lithologies (granite, basalt, sandstone, limestone, schist and ore) and runs in the browser. Its accuracy is measured on a split that holds out entire boreholes, 14 holes for training (4704 windows) and 4 for testing (1344 windows), so no window from a test hole is seen in training; random patch splits inflate core-logging accuracy by letting near-identical patches of one hole appear on both sides. On this split (Fig. 1) the network reaches 99.4% accuracy, against 92.9% for a nearest-centroid classifier on colour and texture features evaluated on the same windows, a margin of 6.5 percentage points. Its per-class recall lies between 0.986 (sandstone) and 1.000 (basalt, limestone).

III. REAL CORE AND THE SIM-TO-REAL GAP

The real core comes from the open drill-core image dataset DCID [6], in its seven-class form DCID-7 (red sandstone,

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Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_CoreLog; the tool is available at <https://corelog.fasl-work.com>.

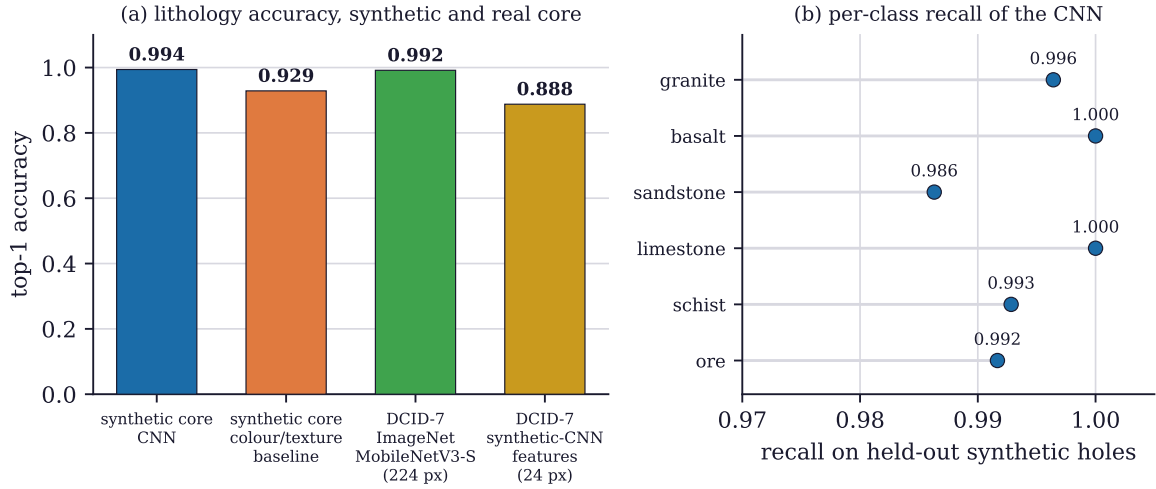


Fig. 1. Lithology classification accuracy. **(a)** On synthetic core the convolutional network reaches 99.4% against 92.9% for a colour-and-texture baseline, on a split that holds out entire boreholes. On real DCID-7 core, a linear head over frozen ImageNet MobileNetV3-Small features (224-pixel input) reaches 99.2% over seven lithologies, and a linear head over the 64-dimensional features of the synthetic-trained network (24-pixel input) reaches 88.8%. **(b)** Per-class recall of the network on the held-out synthetic holes: 0.986 to 1.000 across the six lithologies.

light sandstone, gray siltstone, mudstone, granite, basalt and marble): a class-balanced subset of the dataset’s own train and test split, with perceptual-hash deduplication across the split (2062 training and 829 test images). Two linear classification heads are trained on the real training images over frozen features (Fig. 1a). Over the features of an ImageNet-pretrained MobileNetV3-Small [3] at 224-pixel input, the head reaches 99.2% top-1 accuracy on the test images (macro-F1 0.992). Over the 64-dimensional features of the network trained on synthetic core, with each image reduced to the 24-pixel window that network reads, the head reaches 88.8% (macro-F1 0.887). With permuted training labels the MobileNetV3-Small head falls to chance (0.139, against $1/7 = 0.143$). Real core is therefore learnable to high accuracy from real labels. The 10.4-point difference between the two heads combines two changes, the backbone and the input resolution, and the runs do not separate them, so it bounds the combined effect rather than measuring the cost of synthetic training alone.

IV. DETECTING OUT-OF-DISTRIBUTION CORE

The guard against silent sim-to-real error is an out-of-distribution detector (Fig. 2). Nine detectors are benchmarked on separating real DCID-7 core (out of distribution) from held-out synthetic core (in distribution), with a control against a trivial cue: both streams are reduced to the same 24-pixel window the classifier reads, so no detector can separate them by raw resolution or sharpness alone. The distance detectors work in a network’s feature space, as the Mahalanobis distance to the nearest class mean under a shared covariance [8] or as the distance to the k -th nearest neighbour ($k = 10$) among normalised training features [10]. In the frozen ImageNet features of MobileNetV3-Small and ResNet-18 [2], with the 24-pixel windows upsampled to the backbones’ input size, they reach AUPR 0.976 to 0.999; the best, the Mahalanobis distance in MobileNetV3-Small features, reaches 0.999 (AUROC 0.9995) and flags 0.2% of the synthetic windows at 95% recall of

the real ones. In the classifier’s own 64-dimensional features they reach 0.92 (Mahalanobis: AUROC 0.95, with 28% of synthetic windows flagged at 95% recall); this detector reuses the classifier’s features and is the one the tool runs. The maximum softmax probability [7] and the energy score [9] of the classifier reach 0.83 and 0.74. The reconstruction error of a patch autoencoder trained on synthetic core fails: its AUROC of 0.31 means that real core reconstructs with lower error than held-out synthetic core, so the score ranks the two populations in the wrong order, and its AUPR of 0.28 lies below the chance level of 0.38. To detect that incoming core is unlike the training data, distance in a feature space works on this shift and reconstruction error does not.

V. DISCUSSION AND SCOPE

On synthetic core the classifier reaches 99.4% and exceeds the colour-and-texture baseline; on real core a linear head reaches 99.2% over ImageNet features and 88.8% over the features of the synthetic-trained network; and the real core is separated from the synthetic training core at AUPR up to 0.999 in a feature space, while reconstruction error fails. The measurements have limits. The six-class accuracy is measured on procedurally generated core, which allows controlled evaluation but is not a photograph of real rock. The real-core numbers come from one open dataset, and a deployment would need labels from the target operation. The two real-core heads differ in backbone and input resolution, so their difference is not a clean measure of the sim-to-real gap. The out-of-distribution benchmark tests one large shift, from synthetic to photographed core; smaller shifts between real operations, which motivate the detector, are not tested here.

VI. CONCLUSION

CoreLog Vision classifies drill-core lithology from tray imagery, at 99.4% on held-out synthetic boreholes (against

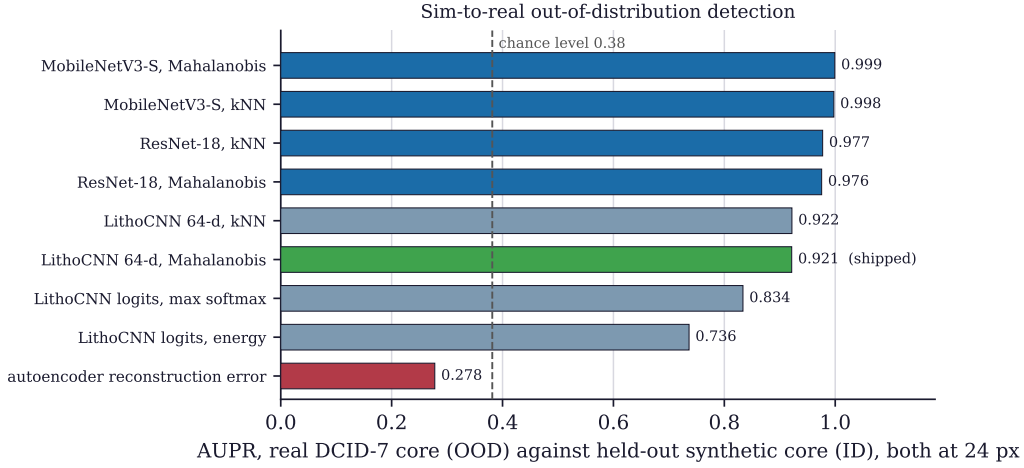


Fig. 2. Sim-to-real out-of-distribution detection: nine detectors separating real DCID-7 core (out of distribution, 829 windows) from held-out synthetic core (in distribution, 1344 windows), scored by the area under the precision-recall curve (AUPR). Both streams are reduced to the same 24-pixel window. The dashed line is the chance level, the out-of-distribution fraction 0.38. Distance detectors in a network’s feature space reach 0.92 to 0.999, the softmax-based scores 0.74 to 0.83, and the autoencoder’s reconstruction error falls below chance (0.28).

92.9% for a colour-and-texture baseline) and at 99.2% on real DCID-7 core with a linear head over ImageNet features. Of nine out-of-distribution detectors, distance detectors in a network’s feature space separate real from synthetic core at AUPR 0.92 to 0.999, and the reconstruction error of an autoencoder ranks them in the wrong order. A feature-space distance that reuses the classifier’s own features is enough to flag this shift.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry the lithology-classifier metrics (`cl-learned.json`: the network and baseline accuracy on the borehole-grouped split and the confusion matrix) and the out-of-distribution benchmark (`ood-bench.json`: the nine detectors, the resolution control, the label permutation null and the two real-core heads on DCID-7). They are produced by `train_litho.py` and `ood_bench.py` under `data-pipeline/pipeline/science/`; `fetch_dcid.py` in the same folder fetches the DCID-7 subset. The figure script under `manuscripts/corelog-vision/figures/` reads `cl.json` in the data folder beside it, a copy of those artifacts. The models are trained with PyTorch and exported to ONNX for the browser.

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